R94/A | Fixed target36 vs R89B

Native FP64 winners; same unique full18 VAL checkpoint on TRAIN/VAL/TEST.

Split	Winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	461	92	655	187	0.5536
VAL	4625	16	4	40	9	0.3459
TEST	4598	22	1	24	12	0.4785

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	-0.445542	22432	28928	0.001287707	0.014429
VAL	-0.249383	1329	1531	0.000903166	0.008845
TEST	-0.129630	1343	1448	0.000615993	0.008485

R89B main / R84A secondary historical; A is the matched random-donor control.

Original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35; dropout .1.

Early trainable inter-node fusion, not R50 pool-first readout. Improvement remains unproven.

16-task donor input coverage, not16-task pretraining; TSP/ATSP raw paths unchanged.

External donor pretraining budget unmatched. No decoder/routes/new cost labels/MTL.

R94/A | Fixed target36 vs R84A

Native FP64 winners; same unique full18 VAL checkpoint on TRAIN/VAL/TEST.

Split	Winners	Correct	Rescued	Lost	New false	Recall (%)
TRAIN	83279	461	240	76	256	0.5536
VAL	4625	16	9	4	16	0.3459
TEST	4598	22	13	4	17	0.4785

Split	Top1 delta pp	Corrected	Harmed	Net cost delta	Regret delta pp
TRAIN	0.001715	23763	23738	0.000203724	0.002865
VAL	-0.086420	1318	1388	0.000585972	0.006027
TEST	0.054321	1333	1289	0.000243401	0.003284

R89B main / R84A secondary historical; A is the matched random-donor control.

Original R77 CE .35 / comparison .35 / risk .02 / R-Drop .35; dropout .1.

Early trainable inter-node fusion, not R50 pool-first readout. Improvement remains unproven.

16-task donor input coverage, not16-task pretraining; TSP/ATSP raw paths unchanged.

External donor pretraining budget unmatched. No decoder/routes/new cost labels/MTL.

R94/A | A Inference Timing

First128 locked TEST inputs/task; synchronized GPU wall time, three repetitions; saved evidence only.

Problem	Batch	Donor + selector ms / instance	Cache-hit selector ms / instance
TSP	128	0.403637	0.393278
CVRP	128	0.488799	0.488662
ATSP	128	0.071193	0.070898
OVRP	128	0.073456	0.072977
VRPB	128	0.073150	0.072927
VRPL	128	0.073301	0.072990
VRPTW	128	0.073165	0.073401
OVRPTW	128	0.073171	0.073107
OVRPB	128	0.073385	0.072881
OVRPL	128	0.073080	0.073038
VRPBL	128	0.073631	0.073041
VRPBTW	128	0.072915	0.072817
VRPLTW	128	0.073276	0.073000
OVRPBL	128	0.072864	0.072992
OVRPBTW	128	0.073097	0.072877
OVRPLTW	128	0.073228	0.073196
VRPBLTW	128	0.072942	0.072968
OVRPBLTW	128	0.073125	0.073002

Inputs already on GPU. Excludes dataset I/O and one-time feature-cache preparation.

Cached-feature selector latency is NOT full-system latency; donor+selector is not I/O-inclusive E2E.

TSP/ATSP have unchanged raw paths and no donor; external pretraining budget is unmatched.

Device: cuda:0